Using Deep Learning for Classifying Ship Trajectories

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***Abstract*— In this paper we demonstrate how deep learning can be applied to the field of sea surveillance by classifying ship types from their trajectories. Commercial ships using AIS continually report information such as their ship type, e.g. fishing or cargo ship. A problem with AIS information however is that it can easily be modified and therefore deliberately or accidentally incorrect. In an attempt to address this we use a 1100 hours long AIS data set to teach 16 different neural networks to classify ships using only motion trajectories and without relying on the reported ship type. We also test three baseline methods using a more conventional 1-nearest neighbor approach. The evaluation showed that the best performing classifier was the one based on deep learning.**

Keywords—deep learning; neural networks; AIS; classification; supervised learning; machine learning; sea surveillance; trajectories

I. INTRODUCTION

Any vessel heavier than 300 tons or categorized as a passenger ship, and subject to international rules, is required to have an AIS (Automatic Identification System) transponder on-board. This transponder sends data such as the ship's name, location, destination, type, speed, course and more. Today civilian and military authorities worldwide access the information as a complement to other sources in systems for maritime surveillance, where it is used to track ships and find anomalies.

A problem with AIS data is that it can be easily modified and therefore, deliberately or accidentally, be incorrect. For instance, a fishing ship can disguise itself by changing the ship type setting of the AIS transponder to transmit cargo ship, this is sometimes called the false ship effect. Even though the AIS transponder can simply be shut off in order to hide one's activities from the AIS network, radars and other systems may still detect the ship making one's actions possibly even more suspect. Therefore, identifying these false ship types can be of great help in stopping activities such as illegal fishing, smuggling and finding disguised military vessels.

The AIS kinematic data for each ship forms a multivariate time series, and determining its ship type thus amounts to sequence classification. Three common techniques for sequence classification are based on features, sequence metrics, and models [1]. Examples on these three are neural networks, k-nearest neighbor and HMM (Hidden Markov Models) respectively.

Different types of ships have different dynamical behaviors, for example cargo ships tend to maneuver more smoothly than fishing vessels. Analyzing all incoming data in enough detail can be difficult and even impossible for human operators due to limited cognitive capabilities. Using a computer program to aid the human operator can be beneficial but designing the algorithms and deciding what features are important can be quite challenging. In the remainder of this section we give an introduction to different techniques for trajectory classification.

In [2] vessel trajectories are analyzed using a machine learning method based on a HMM. They manage to correctly classify fishing activity with respect to expert labeled data with an accuracy of 83-97%, the accuracy varies based on the fishing method. It is intended that this model is to include the techniques in the Global Fishing Watch which provides a sea surveillance application for monitoring fishing activity worldwide, with special interest in illegal fishing[3].

Conformal 1-nearest neighbor (1NN) is proven successful in finding ship anomalies in [4]. Hausdorff distance is used to determine the dissimilarity between trajectories and a single parameter, the similarity threshold, controls the false alarm rate making the method easy to tune.

Deep learning is achieved when machine learning models build a hierarchical representation of the input data. The most common way to do this is by using deep neural networks. These networks consist of several layers of neurons with each layer having an arbitrary number of neurons. As the input data travel deeper into the network it becomes more and more abstract. The final layers in a typical classification neural network do not abstract the data any further but instead serve as a classifier using the feature rich data representation from the previous layers as input. For more information about neural networks and deep learning see [5][6].

In [7][8] neural networks are used to determine anomalous ship behavior as well as predicting future ship location around ports. The network is trained using an associative learning algorithm based on the Hebbian learning rule which takes inspiration from how our own brains learn [9]. This research shows that it is possible to some extent learn ship behavior using neural networks.

Neural networks have successfully been applied to different time series problems outside of sea surveillance in [10][12][13]. In [10] a Multi Channels Deep Convolutional Neural Network

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(MC-DCNN) was shown to perform better than a traditional 1NN when using both Euclidean and DTW distance [11]. Another example of neural networks in sequence classification is found in [12] where the authors argue that important features can be missed when only looking at a single time scale and therefore propose a neural network which works on several time scales. Their Multi-scale Convolutional Neural Network (MCNN) achieves state of the art performance in many time series data sets. In [13] it is found that a Fully Convolutional Network (FCN) outperforms both the conventional DTW and other deep learning methods.

An unusual approach in classifying motion trajectories is shown in [14] where a Convolutional Neural Network (CNN) is used to classify hand drawn gestures meant to convey actions such as dance, run, fly and spin. The unusual part is how they completely remove the time dimension and process the motion trajectories as the resulting image drawn by the motion.

II. METHOD

Several deep learning methods, each with multiple configurations were trained and then validated using an unseen part of the data set. All neural networks use ReLu (Rectifying Linear units) as activation functions and have a softmax layer at the end in order to output probabilities for each ship type[5][6][15]. A baseline was formed using a more conventional 1NN approach. The best performing baseline and deep learning method is compared in the result section.

The AIS data have been captured around the Nordic countries and include roughly 1100 hours of recorded ship traffic data. Four ship types, as reported by the AIS, were extracted for the experiments, namely fishing, passenger, cargo and tanker. The rest of the ship types were not used due to much lower occurrence, see Fig. 1.

A. Filtering

The original data was interpolated into regular time intervals, projected from latitudes and longitudes to Cartesian coordinates, standardized and split into one hour sub trajectories. Ships with too low median speeds were also removed in order to limit the task to moving ships only. Ships with too few data points and ships having time or distance violations were also removed.

B. Baseline

A baseline using three slightly different 1NN approaches were used to compare the deep learning methods with something that is more easily understood. These three baseline methods differ in the distance metrics used, the first uses Euclidian distance to calculate the similarity between trajectories. Euclidean distance was chosen as it is simple and commonly used in sequence classification.

The second baseline method uses Hausdorff distance, this was selected due to its previous success in a related domain, namely anomaly detection [4]. The Hausdorff distance is calculated by first finding all the shortest distances for each point on the first trajectory to the second trajectory and then taking the longest of the set of distance to be the Hausdorff distance. The advantage of using Hausdorff compared to Euclidean distance is that Hausdorff promotes similar shapes better than the Euclidean which instead promotes average separation of the trajectory as a whole.

The third and final distance function is a version of Hausdorff, here named semi-Hausdorff, meant to speed up calculation and include the temporal dimension. This distance metric leverages the fact that all trajectories have the same length and checks the distance between each pair of points in both trajectories in temporal order. Similar to Hausdorff, the longest of the distances is used as the similarity metric.

To avoid having to calculate the distance function against all ships in the training set for every ship in the validation set, a KD-tree in two dimensions was used to shorten the search. The 20 nearest trajectories by median position were then retrieved from a database containing the training data. By doing this computation time was cut by a factor of 1000.

C. Multilayer perceptron

The Multilayer Perceptron (MLP) is a common deep learning classification method. The simple architecture can be applied to almost any classification problem. Four MLP networks with different inputs are used.

- *MLP_Mxy,* using only the median position as input.
- *MLP_xy,* using only position as input
- *MLP_xys,* using position and speed as input
- *MLP_pre_features,* using custom predefined features as input.

The first three MLP have 2048 neurons in the first hidden layer, 1024 neurons in the second hidden layer and 512 neurons in the third hidden layer. After each hidden layer a 25% dropout

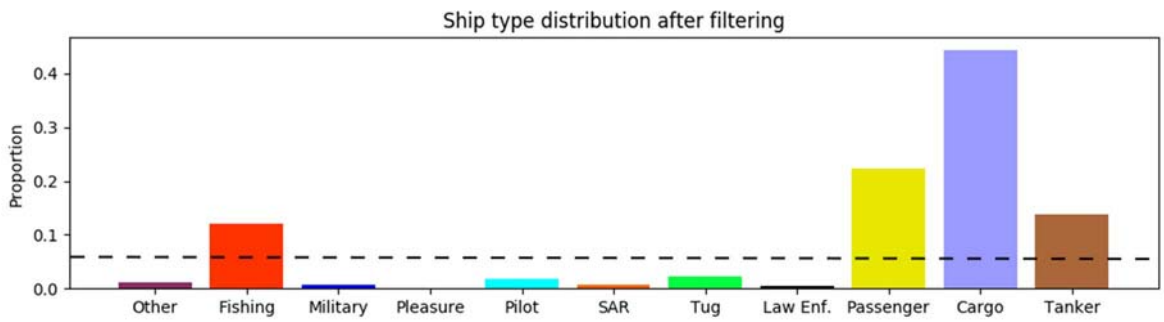

Fig. 1 The ship type distribution of the data set after filtering has been done.

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is made to help generalization. The final layer is the output layer which gives the final estimated confidence levels for each class.

When good key features are already known for a given data set it may be better to provide these than to rely on the network learning its own feature representation. In *MLP_pre_features* predefined features are calculated before entering the neural network. The MLP now works only as classifier. According to [16] fishing ships can be detected by analyzing the frequency used when a ship changes course and speed. It would therefore be a good idea to input a few of the dominating frequencies together with corresponding power for both speed and course. This is done by performing an FFT (Fast Fourier Transform) on each dimension and extracting the 5 frequencies of highest power. Other valuable features included in *MLP_pre_features* are the last known position, mean speed and max speed. The network consists of two hidden layers with 512 neurons in the first hidden layer and 256 neurons in the second. A 25% dropout is also applied in order to help generalization.

D. Multi channels deep convolutional neural network

Using a MC-DCNN as in [10] allows us to perform feature extraction on each channel independently, where a channel being a single time series e.g. course or speed. Each channel in the neural network is structured with two consecutive sections consisting of a convolutional filter layer and a max pooling layer. In Fig. 2 the design of the network can be seen. The filter layer has a kernel size of 5 time steps with 156 feature maps in each layer. The max pooling layer has a kernel size of 4 and a stride of 4. The channels are then merged into a two layered classification network. The first layer is a fully connected layer of 1024 neurons followed by another fully connected hidden layer of 512 neurons. The fully connected classification network also use a 25% dropout in order to help generalization. Two MC-DCNN networks with different input are used.

- *MC_DCNN_Nxy_Mxy,* using normalized position and median position as input.
- *MC_DCNN_sc_Nxy_Mxys* using speed, course and normalized position as input together with the median of both position and speed.

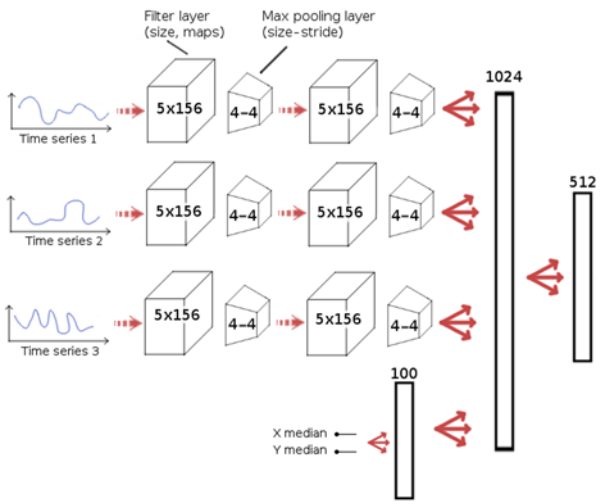

Fig. 2 The DC-MCNN network architecture with three time series and XY median as inputs.

E. Multi-scale convolutional neural network

Some important features which distinguish different ship types apart might lie at different time scales, it is therefore important to have a model that can extract features at multiple time scales. Unlike in [12] where their MCNN only use univariate data, we use multivariate data. To deal with this, the MCNN structure is applied on each time series separately after which all channels are merged similarly to how it is done in MC-DCNN.

Each time series uses three time scales. These three new time series are generated by using max pooling with different strides which in turn compress the original time series. The multi-scale part of the model architecture used in this work can be seen in figure Fig. 3.

- *MCNN_xy,* using only the position as input.
- *MCNN_xysc,* using speed, course and position as input.
- *MCNN_Nxy_Mxy,* using normalized position and median position as input.
- *MCNN_sc_Nxy_Mxy,* using speed, course and normalized position as input together with the median position.

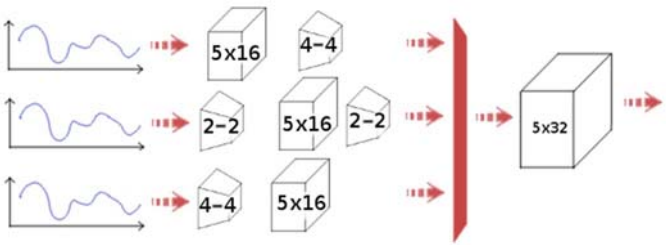

Fig. 3 Multi-scale part of the MCNN architecture showing a single channel.

F. Image based convolutional neural network

The use of Convolutional Neural Networks (CNN) on images have yielded great results in a wide variety of applications and as shown in [14], it is also applicable when using motion trajectories. A CNN approach is therefore evaluated on images of ship trajectories. Each trajectory is drawn onto a 128 by 128 pixels grey scale image. The intensity of the segments is proportional to the speed of the ship at that segment. For all configurations except one, the trajectories are normalized to fit to the fixed image. For the last configuration a global normalization factor is used, this makes distances comparable between samples but also makes some trajectories not fully fit to the image size. Four design configurations in line with [13] are used and the network architecture can be seen in Fig. 4.

- *IMAGE_Nxy,* using normalized position as input.
- *IMAGE_s_GNxy,* using speed and a globally normalized position as input.
- *IMAGE_s_Nxy,* using speed and normalized position as input.
- *IMAGE_s_Nxy_Mxy,* using speed and normalized position as input together with the median position.

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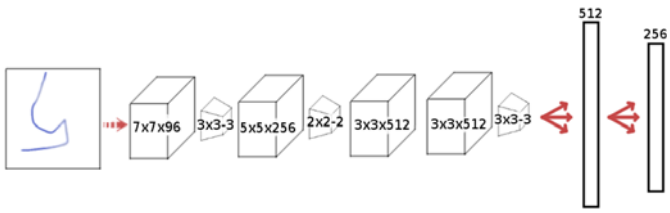

Fig. 4 Network architecture of the image based CNN with the XY spatial trajectory and speed heat-map as inputs.

G. Fully convolutional network

The FCN architecture was shown to perform well using univariate time series in [13]. The same architecture can be used if 2D instead of 1D kernels are used during convolution steps. This method resembles a CNN used on images but with each row representing a time series with equal time steps. Three feature layers are used with batch normalization layers in between, these layers normalize the feature maps and keep the values in the network from getting too big or to diminish. This network does not have any classification section at the end other than the final softmax output layer; this is in line with [13]. Only one configuration of the FCN is used.

- *FCN_xysc*, using speed, course and position.

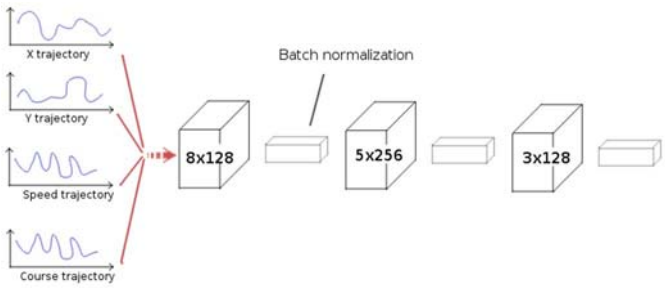

Fig. 5 Network architecture of FCN using batch normalization.

H. Ensemble

All mentioned neural networks have their own feature representation after training is complete. Some architectures are better than others on some parts of the data due to how the features are extracted. By using an ensemble network which include multiple already trained neural networks we can utilize this difference. The listed architectures below were chosen to be part of the ensemble in hope that they all have a varied and rich feature extraction compared to each other.

- *IMAGE_s_Nxy_Mxy*
- *MC_DCNN_sc_Nxy_Mxys*
- *FCN_xysc*
- *MLP_xys*
- *MLP_Mxy*
- *MLP_pre_features*

The final layer for each pre-trained neural network is omitted and the rest of the network is attached to a classification network as shown in figure Fig. 6. This is done because we are only interested in the feature extraction for each model.

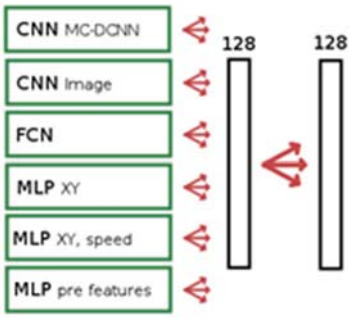

Fig. 6 Network architecture of the ensemble using six previously learned networks.

III. Implementation

A categorical cross entropy loss function together with the Adam optimizer with learning rate at 0.001 was used when training the neural networks. To prevent overfitting on the training data early stopping was used. The training was done on a laptop with an Nvidia Quadro M2000M which for all neural networks took two weeks all together. The code was written using the Keras python library with Theano backend[17][18].

The ensemble network is different from the others in that the included networks are trained separately using an oversampled data set. This is meant to balance number of samples for all ship types and to prevent biases during feature extraction. The oversampling is limited to three times the original size. The ensemble network is trained as normal with the real ship type distribution after all the sub networks have completed training.

All the 18 methods were evaluated based on the classification accuracy. This is defined as the percentage of accurately predicted ship types. A totally random classifier would score 25% and a classifier predicting only the dominant ship type in this data set would score roughly 45%.

IV. Results

An overview of the classification accuracy for each method can be seen in Fig. 7. The best performing method is the deep learning ensemble method with a classification rate of 81%. The baseline methods are clustered to the center of the graph with the Euclidian distance scoring the highest out of the three. Worth noticing is that the MLP using only the median position as input scores higher than that of the Hausdorff baseline.

The MCNN methods, having the most complex architecture, are at the bottom of the graph as opposed the simpler MLP and FCN methods which are found at the top. The larger and more complicated networks proved hard to train effectively and would quickly suffer from overfitting.

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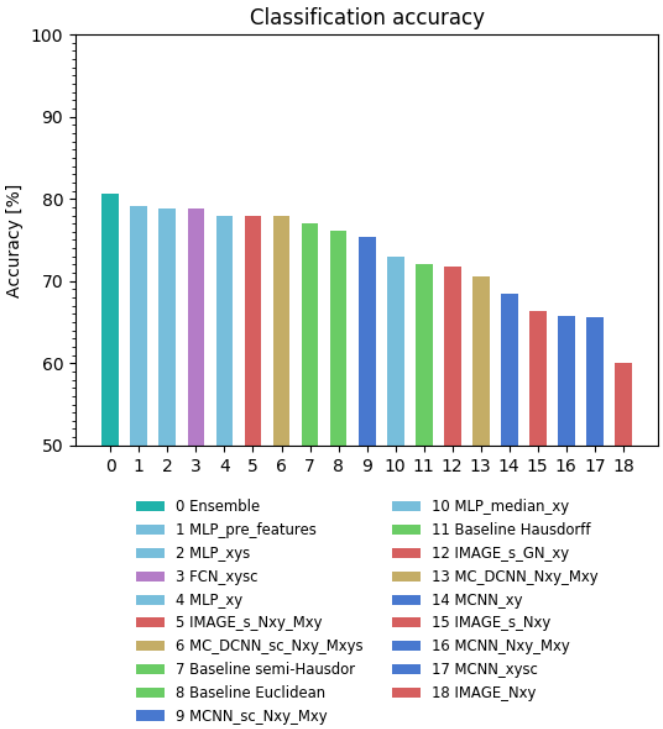

Fig. 7 Overview of classification accuracy for all methods. Colors indicate method family and the error bars show the spread.

The results in Fig. 7 can be a bit deceiving as they show the classification accuracy but ignore the confidence of the prediction. By disregarding the confidence, we are forcing a classification at the cost of a higher false positive rate. The risk of not detecting a false ship has to be put in comparison to the number of false positives. It can be argued that for a human operator it would be better to only present a few confident classifications rather than many uncertain ones. Therefore, tuning the false positive rate is of great importance. By looking at Fig. 8 and Fig. 9 we can see how the best performing 1NN method compares to the best performing deep learning method. Analysing the 1NN further we see that even though the distance threshold is lowered the overall false positive rate decrease only slightly. This indicates an inability to efficiently use motion trajectory behaviour as a means to evaluate the ship type and could mean that the distance function has difficulty in comparing trajectory shapes. The 1NN methods are reducing the problem to points on a map. This hypothesis is also enforced by the fact that the MLP network using only median position as input manages to score roughly the same as the three 1NN baselines. The baseline methods might need trajectories longer than 1 hour and more data in order to make historical trajectories close enough for the distance metric to work properly.

Looking at Fig. 9 we see that the deep learning method unlike the baseline method benefits from the lower prediction rates. The overall false positive rate at 50% classification rate decreases to become an astonishing 5% as compared to the 24% for the baseline. Also, worth noticing is the steep decline in false positive rate for the fishing type which is most likely due to it being easier to distinguish fishing ships during fishing activity compared to when traveling in sea lanes or in and out from ports.

In sea lanes and close to ports the overall distances between ships are on average smaller than on open sea. This means that while lowering the distance threshold for the 1NN method could lead to better predictions it can also make it worse. This can be seen with fishing ships, in fishing waters they are easy to categorize while in sea lanes and ports it is much more difficult. This could explain the sudden rise in false positive rate for fishing ships in Fig. 8. The same explanation can however not be applied in Fig. 9 where the cargo ship type increases until it finally decreases. This behaviour is harder to analyse due to the black box nature of neural networks but could be that the network prioritizes cargo over tanker ships more and more.

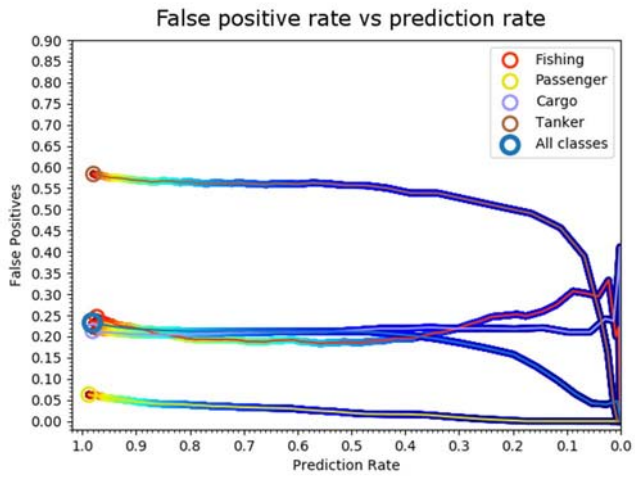

Fig. 8 False positive rate as a function of classification rate for the semi-Hausdorff baseline method. The color of each line show how the distance threshold decrease with the prediction rate. Red color indicate a high threshold and dark blue a low threshold which only include small distances.

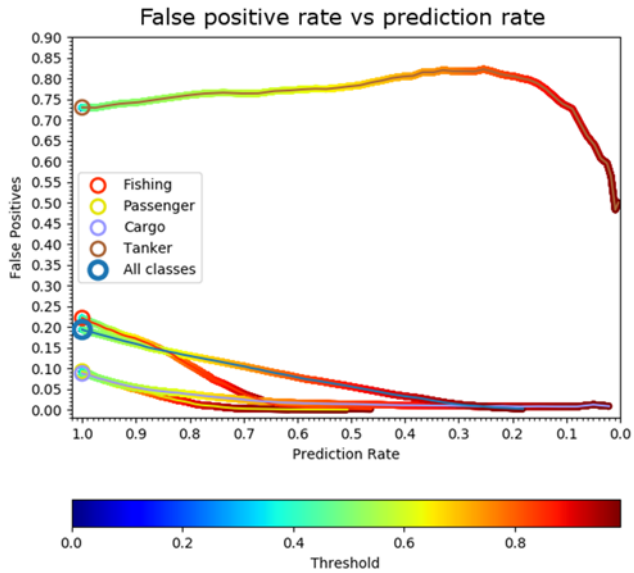

Fig. 9 False positive rate as a function of prediction rate for the deep learning ensemble method. The color of each line show how the distance threshold increase with the prediction rate. Confident predictions have a value close to one and by increasing the threshold we only use confident prediction.

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Both the ensemble and baseline methods performed poorly on the tanker type, the reason for this becomes clear when looking at the confusion matrices for the two methods found in Fig. 10 and Fig. 11. We see that both methods had difficulties in distinguishing the tanker and cargo type, this is understandable given how similar the two types are to each other. The baseline method is naturally biased to predict cargo more than to predict tanker due to the ship type distribution. Whereas the deep learning method learns the ship type distribution and can then when unsure predict cargo rather than tanker in order to increase performance. It can be argued that because the two types are so similar it makes little to no difference in using a common ship type. Tests with tanker and cargo as a single ship type called transport showed overall classification accuracies of over 90 %. It was however decided that in this paper, to better understand how the algorithms work, that these two ship types remain separate.

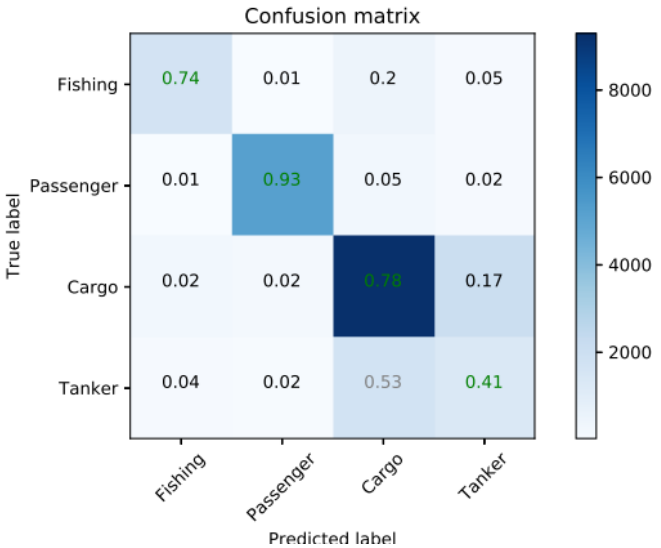

Fig. 10 Confusion matrix for the semi-Hausdorff baseline.

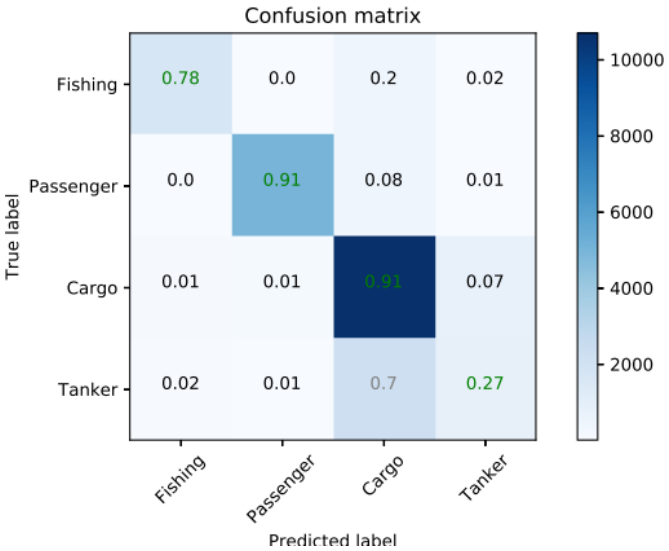

Fig. 11 Confusion matrix for the ensemble method.

The confusion matrices further show that the fishing and passenger ships are also often misclassified as cargo ships. Again this is due to the higher number of the latter, and also that portions of their trajectories do not look too different from those of cargo ships traveling in sea lanes.

V. Conclusion

In this paper it is shown that deep neural networks are capable of learning ship specific features and manage to classify a set of ship types better than our 1NN baseline methods. The advantages of the deep learning methods appear most evident when taking classification confidence into account. For example, by only considering the 50% of the classifications that have the highest confidence, the ensemble outperforms the baseline by a large margin.

A. Future work

A common strategy for sequence classification is by using recurrent neural networks [19]. This was not tested due to lack of time, but it would be interesting to see how well it performs. A key feature which would be very beneficial for AIS time series is that the input data does not have to be a fixed length. This makes it also possible to use trajectories of arbitrary lengths.

To further enhance the performance of the classification algorithms it would be a good idea to include meta-data such as point of departure or destination which is included in the AIS data. Even auxiliary information such as time of day and weather conditions could help in improving performance.

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References

[1] Xing, Z., Pei, J., & Keogh, E.. “A brief survey on sequence classification”, ACM Sigkdd Explorations Newsletter, 12(1), 40-48, 2010.

[2] de Souza, E. N., Boerder, K., Matwin, S., & Worm, B. “Improving fishing pattern detection from satellite ais using data mining and machine learning”, PloS one, 11(7), 2016.

[3] Website http://globalfishingwatch.org/, [accessed jan 2018].

[4] R. Laxhammar and G. Falkman, “Sequential conformal anomaly detection in trajectories based on hausdorff distance”, Information Fusion 14th International Conference, Chicago, USA, pp. 1-8, 2011.

[5] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016, http://www.deeplearningbook.org, [accessed jan 2018].

[6] M. A. Nielsen, Neural Networks and Deep Learning. Determination Press, 2015, http://neuralnetworksanddeeplearning.com/, [accessed jan 2018].

[7] Bomberger, N. A., Rhodes, B. J., Seibert, M., & Waxman, A. M.. “Associative learning of vessel motion patterns for maritime situation awareness”, Information Fusion 9th International Conference, Florence, Italy, pp. 1-8. 2006.

[8] Rhodes, B. J., Bomberger, N. A., & Zandipour, M.. “Probabilistic associative learning of vessel motion patterns at multiple spatial scales for maritime situation awareness”, Information Fusion 10th International Conference, Quebec, Canada, pp. 1-8, 2007.

[9] Hebb, Donald O. "The organization of behavior: A neuropsychological theory.", 1949.

[10] Zheng, Y., Liu, Q., Chen, E., Ge, Y., & Zhao, J. L. ”Time series classification using multi-channels deep convolutional neural networks”,

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International Conference on Web-Age Information Management, Macau, China, pp 298-310, ,2014.

[11] Górecki, T., & Łuczak, M., "Multivariate time series classification with parametric derivative dynamic time warping.", Expert Systems with Applications, 42(5), pp. 2305-2312, 2015.

[12] Cui, Z., Chen, W., & Chen, Y. "Multi-scale convolutional neural networks for time series classification", arXiv:1603.06995, 2016.

[13] Wang, Z., Yan, W., & Oates, "Time series classification from scratch with deep neural networks: A strong baseline", In Neural Networks (IJCNN), 2017 International Joint Conference, Anchorage, USA , pp. 1578-1585, 2017.

[14] Singh, K. Y., Davis, N., Hsiao, C. P., Jacob, M., Patel, K., & Magerko, B. "Recognizing Actions in Motion Trajectories Using Deep Neural Networks", 12th Artificial Intelligence and Interactive Digital Entertainment Conference, Palo Alto, California, pp 211-217, 2016.

[15] Krizhevsky, A., Sutskever, I., & Hinton, G. E., "Imagenet classification with deep convolutional neural networks.", Advances in neural information processing systems, pp. 1097-1105.

[16] White, Timothy D., et al. "Assessing the effectiveness of a large marine protected area for reef shark conservation", Biological Conservation vol 207, pp 64-71, 2017.

[17] Franois Chollet, Keras, 2015, https://github.com/fchollet/keras, [accessed jan 2018].

[18] Al-Rfou, R., Alain, G., Almahairi, A., Angermueller, C., Bahdanau, D., Ballas, N., ... & Bengio, Y., "Theano: A Python framework for fast computation of mathematical expressions", arXiv preprint arXiv:1605.02688, 472, 473, 2016.

[19] Graves, A. "Supervised sequence labelling with recurrent neural networks" Springer, 2014.

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